

Central Library
Nitte

Max. Marks: 20

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Express the Boolean function $F(A, B, C) = A + \bar{B}C$ as a sum of minterms and $F(X, Y, Z) = (X + Y)(\bar{X} + Z)$ as a product of maxterms.	6	L*3	2	1
	b) State and prove De-Morgan's theorem using the tabular method.	4	L2	2	1
2.	a) Perform conversion on the following numbers, clearly showing the steps.				
	i. $(25.625)_{10} = (?)_2$				
	ii. $(42A.12)_{16} = (?)_{10}$				
	iii. $(1011001.0110111)_2 = (?)_{16}$	6	L3	1	1
	b) Write the difference between Analog and Digital Signals.	4	L2	1	1
Unit – II					
3.	a) Design a combinational logic circuit so that an output is generated indicating when the majority of four inputs are true.	5	L3	2	1
	b) Simplify the given Boolean Expression using the K-map method.				
	$Y(A, B, C, D) = \sum m(0, 1, 3, 7, 8, 9, 11, 15)$	5	L3	2	1
4.	a) For the given incompletely specified Boolean function, find the minimal sum.				
	$F(a, b, c, d) = \sum m(1, 5, 6, 7, 9, 11, 12, 13) + \sum d(0, 3, 4)$	5	L3	2	1
	b) Realize the given Boolean expression using NOR gate only with the help of a logic diagram.				
	$Y(A, B, C) = A\bar{B} + \bar{A}B + AC$	5	L3	2	1
